

Dataset	Description	Mutation type	# Proteins	# Mutants
DMS	High-throughput assays evaluating the functional impact of a wide range of protein mutations	Substitutions	217	2.4M
		Indels	66	0.3M
Clinical	Expert-curated clinical annotations across a wide range of human genes	Substitutions	2,525	63k
		Indels	1,555	3k
Total			3,422	2.7M

Table 1: **ProteinGym datasets summary** ProteinGym includes a large collection of DMS assays and clinical datasets that offer complementary viewpoints when assessing protein fitness. The table reports the number of mutants and unique proteins per dataset (the total being deduped across datasets).

need to rely on labels is even more pronounced when we seek to optimize several anti-correlated properties or when evolution is a poor proxy for the property of interest. Predictions obtained in the zero-shot settings may also be used to augment supervised models [Hsu et al., 2022a]. The two settings require substantially different evaluation frameworks, which we detail in § 4.

4 Evaluation framework

4.1 Zero-shot benchmarks

DMS assays In the zero-shot setting we predict experimental phenotypical measurements from a given assay, without having access to the labels at training time. Due to the often non-linear relationship between protein function and organism fitness [Boucher et al., 2016], the Spearman’s rank correlation coefficient is the most generally appropriate metric for model performance on experimental measurements. We use this metric similarly to previous studies [Hopf et al., 2017, Riesselman et al., 2018, Meier et al., 2021]. However, in situations where DMS measurements exhibit a bimodal profile, rank correlations may not be the optimal choice. Consequently, for these instances, we supplement our performance assessment with additional metrics, namely the Area Under the ROC Curve (AUC), and the Matthews Correlation Coefficient (MCC), which compare model scores with binarized experimental measurements. Furthermore, for certain goals (e.g., optimizing functional properties of designed proteins), it is more important that a model is able to correctly identify the most functional protein variants, rather than properly capture the overall distribution of all assayed variants. Thus, we also calculate the Normalized Discounted Cumulative Gains (NDCG), which up-weights a model if it gives its highest scores to sequences with the highest DMS value. We also calculate Top K Recall, where we select K to be the top 10% of DMS values. To avoid placing too much weight on properties where we have many assays (e.g., thermostability), we first compute each of these metrics within groups of assays that measure similar functions. The final value of the metric is then the average of these averages, giving each functional group equal weight. We refer to the corresponding value as ‘corrected average’.

Clinical datasets For the clinical data, with pathogenic and benign categories, we calculate the areas under the ROC and precision-recall curves. In the substitution dataset, 50% of the labels are in approximately 10% of the proteins. Since clinical labels across genes correspond to underlying pathologies that are very distinct to one another, it is preferable to assess performance on a gene-by-gene basis. We thus compute the average per-gene performance on the substitution benchmark. However, in the case of indels, only about half of the proteins has a pathogenic label (and only 10% have a both pathogenic and benign or pseudocontrol labels), so we compute the total AUC for the full dataset. The problem of calibrating model scores in a principled way across different genes is an open problem; we leave this to future work.

Baselines We implement a diverse set of 50+ zero-shot baselines that may be grouped into alignment-based models, protein language models, inverse folding models and ‘hybrid’ models. Alignment-based models, such as site-independent and EVmutation models [Hopf et al., 2017], DeepSequence [Riesselman et al., 2018], WaveNet [Shin et al., 2021], EVE [Frazer et al., 2021] and GEMME [Laine et al., 2019], are trained on Multiple Sequence Alignments (MSAs). Protein language models are trained on large quantities of unaligned sequences across protein families. They

include UniRep [Alley et al., 2019a], the RITA suite [Hesslow et al., 2022], the ESM1 and ESM2 suite [Rives et al., 2021, Meier et al., 2021, Lin et al., 2023], VESPA [Marquet et al., 2022], the CARP suite [Yang et al., 2023a] and the ProGen2 suite [Nijkamp et al., 2022]. Inverse Folding models learn sequence distributions conditional on an input structure [Ingraham et al., 2019]. We include here ProteinMPNN [Dauparas et al., 2022] which is trained on structures in the PDB, MIF [Yang et al., 2023b] trained on CATH4.2 [Dawson et al., 2016], and ESM-IF1 [Hsu et al., 2022b] which is trained on the PDB and a dataset of AlphaFold2 folded structures. Hybrid models combine the respective strengths of family-specific alignment-based and family-agnostic language models, such as the MSA Transformer [Rao et al., 2021], evotuned UniRep [Alley et al., 2019a], Tranception [Notin et al., 2022a] and TranceptEVE [Notin et al., 2022b].

Because of the variable length of sequences subject to insertion or deletion mutations, alignment-based methods with fixed matrix representations of sequences are unable to score indels. However, profile Hidden Markov Model (HMM) and autoregressive models include explicit or implicit probabilities of indels at each position. Both are trained on homologous sequences recovered with an MSA and expanded to include insertions. The masked-marginals heuristic Meier et al. [2021] used to predict protein fitness with protein language models trained with a masked-language modeling objective (e.g., ESM-1v, MSA Transformer) does not support indels (see Appendix A.4). We thus only report the performance of the following baselines: Tranception [Notin et al., 2022a], TranceptEVE [Notin et al., 2022b], WaveNet [Shin et al., 2021], HMM [Eddy, 2011], ProGen2 [Madani et al., 2020], UniRep [Alley et al., 2019a], RITA [Hesslow et al., 2022] and ProtGPT2 [Ferruz et al., 2022].

For comparisons on clinical benchmarks, we also include unsupervised baselines developed for variant effect prediction in humans, such as SIFT [Ng and Henikoff, 2002], MutPred [Li et al., 2009], LRT [Chun and Fay, 2009], MutationAssessor [Reva et al., 2011], PROVEAN [Choi et al., 2012], PrimateAI [Sundaram et al., 2018] and LIST-S2 [Malhis et al., 2020].

4.2 Supervised benchmarks

DMS assays We leverage the same set of 250+ substitutions and indels DMS assays as for the zero-shot setting. In the supervised setting, greater care should be dedicated to mitigating overfitting risks, as the observations in biological datasets may not be fully independent. For instance, two mutations involving amino acids with similar biochemical properties at the same position will tend to produce similar effects. If we train on one of these mutations and test on the other, we will tend to overestimate our ability to predict the effects of mutants at unseen positions. In order to quantify the ability of each model to extrapolate to unseen positions at training time, we leverage 3 types of cross-validation schemes introduced in Notin et al. [2023]. In the *Random* scheme, each mutation is randomly assigned to one of five different folds. In the *Contiguous* scheme, we split the sequence contiguously along its length, in order to obtain 5 segments of contiguous positions, and assign mutations to each segment based on the position at which it occurs. Lastly, in the *Modulo* scheme, we assign positions to each fold using the modulo operator to obtain 5 folds overall. In all supervised settings, we report both the Spearman’s rank correlation and Mean Squared Error (MSE) between predictions and experimental measurements. A more challenging generalization task would involve learning the relationship between protein representation (sequence, structure, or both) and function using only a handful of proteins, and then extrapolating at inference time to protein families not encountered during training. This setting may be seen as a hybrid between the zero-shot and supervised regimes – closer to zero-shot if we seek to predict different properties across families, and closer to the supervised setting if the properties are similar (eg., predicting the thermostability of proteins with low sequence similarity with the ones in the training set). While this study does not delve into these hybrid scenarios, the DMS assays in ProteinGym can facilitate such analyses.

Clinical datasets Given the restrictions on the number of labels available per gene and the discrepancies between train-validation-test splits across the different supervised baselines, we report test performance on the full set of all available ClinVar labels. We note that this may result in overestimating the performance of supervised methods for which the training data would substantially overlap with the labels considered in our ClinVar set. Further data leakage occurs for models trained on population frequencies, as most ClinVar benign labels are established based on observed frequencies in humans (situation especially evident for our indel dataset where we use frequent variants as pseudocontrols). Interestingly, despite this overfitting risk and as first observed in Frazer et al. [2021], we find that most supervised methods are outperformed by the best unsupervised methods (Fig. 2).

Baselines For the supervised DMS benchmark, we report two suites of baselines. The first suite is comprised of models that take as inputs One-Hot-Encoded (OHE) features. Following the protocol described in Hsu et al. [2022a], we augment the model inputs with predictions from several state-of-the-art zero-shot baselines: DeepSequence [Riesselman et al., 2018], ESM-1v [Meier et al., 2021], MSA Transformer [Rao et al., 2019], Tranception [Notin et al., 2022a] and TranceptEVE [Notin et al., 2022b]. Following prior works from the semi-supervised protein modeling literature [Heinzinger et al., 2019, Dallago et al., 2021], the second suite is formed with baselines that leverage mean-pooled embeddings from several protein language models (ESM-1v, MSA Transformer and Tranception) in lieu of OHE features. We also augment these baselines with zero-shot predictions obtained with the same model used to extract the protein sequence embeddings. Lastly, we include ProteinNPT [Notin et al., 2023], a semi-supervised pseudo-generative architecture which jointly models sequences and labels by performing axial attention [Ho et al., 2019b, Kossen et al., 2022] on input labeled batches. Additional details for the corresponding model architectures are reported in Appendix A.4.2. On the various clinical benchmarks, the above baselines are challenging to train given the low number of labels available per gene. We instead include several supervised baselines that have been specifically developed for variant effects predictions in humans, such as ClinPred [Alirezaie et al., 2018], MetaRNN [Li et al., 2022], BayesDel [Feng, 2017], REVEL [Ioannidis et al., 2016] and PolyPhen-2 [Adzhubei et al., 2010] (full list in A.4.3).

5 Results

5.1 Substitution benchmarks

We follow the experimental protocol described in § 4.1 and report our main results on the zero-shot DMS benchmarks in Table 2, supervised DMS benchmark in Table 3, and combined supervised and unsupervised clinical benchmarks in Fig. 2A. TranceptEVE emerges as the best overall method across the various settings. One of the key objectives of ProteinGym benchmarks is to analyze performance across a wide range of regimes to guide model selection depending on the objectives of the practitioners. To that end we also provide a performance breakdown across MSA depth, mutational depth and taxa where relevant (see Appendix A.5 and supplements). While TranceptEVE tops the ranking across the majority of metrics and settings, GEMME achieves the best performance in several categories, such as assays of viral or non-human eukaryotic proteins, and low and medium depth MSAs. While we report average performance per metric in Table 2, the *distribution* of scores across assays is also insightful. For instance, certain models are heavily penalized in aggregate rankings due to very poor performance on a handful of assays (e.g., ESM-1v), such that looking at the median performance in lieu of the average provides a complementary viewpoint. Furthermore, although most models rank similarly under Spearman and NDCG, some have comparatively better performance in one over the other (Fig. 2B). Superior ranking under NDCG may suggest a model is better at predicting the top end of a score distribution, which may be a desirable feature when using models for design and optimization. Many of the alignment-based methods (e.g. EVmutation, WaveNet) exhibit this behavior (Fig. A1). Models with higher relative Spearman (e.g., ESM-1v and ESM-2) may be more effective for cases where the model needs to learn the full property distribution well, such as with mutation effect prediction. Lastly, in the zero-shot setting, autoregressive protein language models (e.g., Tranception, ProGen2) tend to outperform their masked language modeling counterparts (e.g., ESM models). However, in supervised settings, both types of models provide valuable embeddings for learning. The optimal method depends on the specific situation, as observed in Table 3 and Table A16. The best performance is achieved with the ProteinNPT architecture, demonstrating the value from performing self-attention alternatively across columns (i.e., amino acid tokens and labels) and rows (i.e., protein sequences) to learn a rich representation of the data.

5.2 Indel benchmarks

The zero-shot results for an indel-compatible subset of the models in ProteinGym is shown in Table 4. The Spearman rank correlations are separated by the method used to generate test sequences: unbiased libraries, or model-designed sequences biased towards natural sequences. Model performance exhibits higher variance across assay types, with ProGen2 achieving the highest performance on Library assays (albeit with low performance on designed assays), WaveNet topping the ranking on designed assays (but with low performance on library assays), and TranceptEVE reaching high performance across both. We provide additional indel results in the supervised and clinical settings in Appendix A.5.

Model type	Model name	Spearman	AUC	MCC	NDCG	Recall
Alignment-based	Site-Independent	0.359	0.696	0.286	0.747	0.201
	WaveNet	0.373	0.707	0.294	0.761	0.203
	EVmutation	0.395	0.716	0.305	0.777	0.222
	DeepSequence (ensemble)	0.419	0.729	0.328	0.776	0.226
	EVE (ensemble)	0.439	0.741	0.342	0.783	0.230
	GEMME	0.455	0.749	0.352	0.777	0.211
Protein language	UniRep	0.190	0.605	0.147	0.647	0.139
	CARP (640M)	0.368	0.701	0.285	0.748	0.208
	ESM-1b	0.394	0.719	0.311	0.747	0.203
	ESM-2 (15B)	0.401	0.720	0.314	0.759	0.208
	RITA XL	0.372	0.707	0.293	0.751	0.193
	ESM-1v (ensemble)	0.407	0.723	0.320	0.749	0.211
	ProGen2 XL	0.391	0.717	0.306	0.767	0.199
	VESPA	0.436	0.742	0.346	0.775	0.201
Hybrid	UniRep evotuned	0.347	0.693	0.274	0.739	0.181
	MSA Transformer (ensemble)	0.434	0.738	0.340	0.779	0.224
	Tranception L	0.434	0.739	0.341	0.779	0.220
	TranceptEVE L	0.456	0.751	0.356	0.786	0.230
Inverse Folding	ESM-IF1	0.422	0.730	0.331	0.748	0.223
	MIF-ST	0.401	0.718	0.311	0.766	0.226
	ProteinMPNN	0.258	0.639	0.196	0.713	0.186

Table 2: **Zero-shot substitution DMS benchmark** Corrected average of Spearman’s rank correlation, AUC, MCC, NDCG@10%, and top 10% recall between model scores and experimental measurements on the ProteinGym substitution benchmark.

Model type	Model name	Spearman ($\uparrow$)				MSE ($\downarrow$)			
		Contig.	Mod.	Rand.	Avg.	Contig.	Mod.	Rand.	Avg.
OHE	None	0.064	0.027	0.579	0.224	1.158	1.125	0.898	1.061
	DeepSequence	0.400	0.400	0.521	0.440	0.967	0.940	0.767	0.891
	ESM-1v	0.367	0.368	0.514	0.417	0.977	0.949	0.764	0.897
	MSAT	0.410	0.412	0.536	0.453	0.963	0.934	0.749	0.882
	Tranception	0.419	0.419	0.535	0.458	0.985	0.934	0.766	0.895
	TranceptEVE	0.441	0.440	0.550	0.477	0.953	0.914	0.743	0.870
Embed.	ESM-1v	0.481	0.506	0.639	0.542	0.937	0.861	0.563	0.787
	MSAT	0.525	0.538	0.642	0.568	0.836	0.795	0.573	0.735
	Tranception	0.490	0.526	0.696	0.571	0.972	0.833	0.503	0.769
NPT	ProteinNPT	0.547	0.564	0.730	0.613	0.820	0.771	0.459	0.683

Table 3: **Supervised substitution DMS benchmark**. Corrected average of Spearman’s rank correlation and MSE between model predictions and experimental measurements. MSAT is a shorthand for MSA Transformer.

6 Resources

Codebase A key contribution of our work is the consolidation of the numerous baselines discussed in § 4 in a single open-source GitHub repository. While the main code for the majority of these baselines is publicly available, it often does not support fitness prediction out-of-the-box or, when it does, the codebase does not necessarily provide all the required data processing logic (e.g., pre-processing of MSAs in MSA Transformer) or handle all possible edge cases that may be encountered (e.g., scoring of sequences longer than context size in the ESM suite). Our GitHub repository addresses all of these gaps and provides a consistent interface that will aid in the seamless integration of new baselines as they become available.

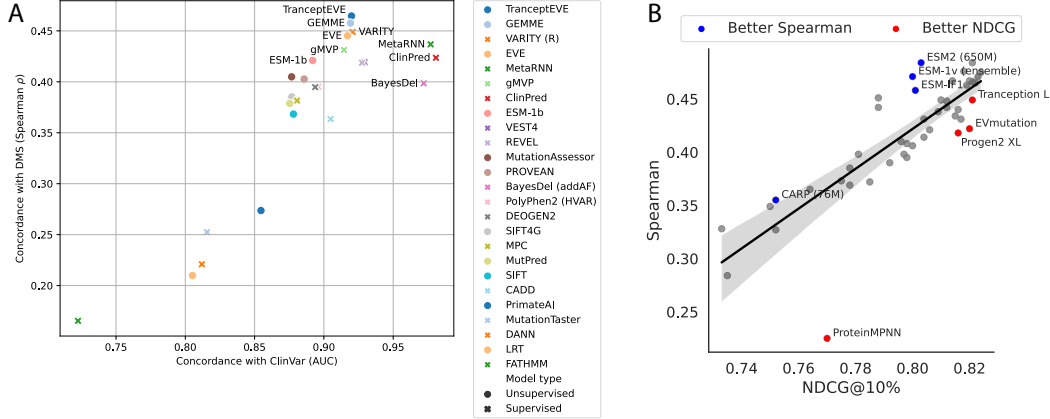


Figure 2: **Comparing baselines across datasets and across performance metrics** (A) Performance estimated against known clinical labels (avg. AUC over genes in ClinVar (x axis)), and DMS assays assessing the clinical effect of variants in humans (avg. Spearman (y axis)). (B) The zero-shot models’ median NDCG@10% (x-axis) against median Spearman (y-axis) on the DMS substitutions.

Model type	Model name	Spearman by DMS type (↑)			AUC (↑) All
		Library	Designed/Natural	All	
Alignment models	HMM	0.373	0.518	0.389	0.744
	WaveNet	0.323	0.597	0.368	0.720
	PROVEAN	0.306	0.585	0.347	0.725
Protein language models	RITA L	0.443	0.519	0.457	0.773
	ProtGPT2	0.185	0.128	0.191	0.620
	ProGen2 M	0.472	0.205	0.465	0.776
Hybrid models	Tranception M	0.395	0.544	0.394	0.733
	Tranception L	0.387	0.563	0.395	0.741
	TranceptEVE M	0.426	0.587	0.424	0.754

Table 4: **Zero-shot indel DMS benchmark** Spearman’s rank correlations and AUC between model scores and experimental measurements.

Processed datasets We also make publicly available all processed datasets used in our various benchmarks in a consistent format, including all DMS assays, model scores, ClinVar/gnomAD datasets, predicted 3D structures and Multiple Sequence Alignments required for training and scoring (see Section A.3.3 for more details).

Website Lastly, we developed a user-friendly website in which all benchmarks are accessible, with functionalities to support drill analyses across various dimensions (e.g., mutational depth, taxa) and exporting capabilities.

7 Conclusion

ProteinGym addresses the lack of large-scale benchmarks for the robust assessment of models developed for protein design and fitness prediction. It facilitates the direct comparison of methods across several dimensions of interest (e.g., MSA depth, mutational depth, taxa), based on different ground truth datasets (e.g., DMS assays vs Clinical annotations), and in different regimes (e.g., zero-shot vs supervised). We expect the ProteinGym benchmarks and the various data assets we publicly release along with them, to be valuable resources for the Machine Learning and Computational Biology communities, and we plan to continue updating the benchmarks as new assays and baselines become available.